Test 2 – Grade 9

Name: _____
Grade / Class: 9

Section A: English Language & Literature

1. Reading Comprehension
Read the passage and answer the questions:

"During winter holidays, Ahmed joined a robotics workshop. He learned to assemble circuits, program basic robots, and collaborate with peers on problem-solving tasks. The workshop emphasized innovation, critical thinking, and practical applications of technology."

a) What did Ahmed learn at the workshop?
b) How did the workshop encourage teamwork?
c) Which skills are highlighted in the passage?
d) Summarize the passage in your own four sentences.

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2. Grammar & Vocabulary

a) Combine using an appropriate conjunction:
"The train was delayed. Passengers remained calm."

b) Identify adjectives and adverbs:
"The young boy quickly solved the complicated puzzle."

c) Correct the error:
"Neither the teacher nor the students knows the answer."

d) Convert to indirect speech:
"She said, 'I will submit my project next Monday.'"

e) Replace the underlined word with a synonym:
"The view from the hill was magnificent."

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3. Writing

Write a short essay (120–150 words) on:
“Advantages and Disadvantages of Online Learning”
Include examples and personal opinion.

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4. Literature Analysis

a) Discuss the central conflict in a novel or story you studied.
b) Identify the simile in the sentence:
"Her smile was as bright as the morning sun."
c) Analyze the motivations of a secondary character from your selected reading.

Section B: Mathematics

1. Algebra

a) Solve for x:
8x−20=3x+158x - 20 = 3x + 158x−20=3x+15

b) Factorize:
x2−13x+42x^2 - 13x + 42x2−13x+42

c) Simplify:
5a2b−6ab2+2a2b5a^2b - 6ab^2 + 2a^2b5a2b−6ab2+2a2b

d) Solve simultaneously:
3x+4y=143x + 4y = 143x+4y=14, 2x−y=32x - y = 32x−y=3

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2. Geometry & Mensuration

a) Find the area of a parallelogram with base 10 cm and height 8 cm.
b) A cylinder has radius 6 cm and height 20 cm. Find its volume.
c) Find the midpoint of the line segment joining points (4, -5) and (-2, 7).

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3. Trigonometry

a) Find cos θ if sin θ = 0.8 and θ is acute.
b) Solve: 1+tan2x=sec2x1 + tan^2x = sec^2x1+tan2x=sec2x
c) A flagpole casts a shadow 12 m long. If the angle of elevation is 45°, find the height of the flagpole.

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4. Ratio, Proportion & Percentage

a) A bicycle costs $250. It is sold at 10% discount. Find selling price.
b) 7 pens cost $21. Find the price of 15 pens.
c) A city's population grew from 80,000 to 88,000. Calculate the percentage increase.

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5. Statistics & Probability

a) Marks of 10 students: 10, 12, 15, 10, 14, 13, 12, 15, 11, 14. Find mean, median, mode.
b) Construct a frequency table for the above data.
c) A bag has 3 red, 4 blue, and 5 black balls. Probability of picking a blue ball?

Section C: Science

1. Biology

a) Describe the human respiratory system.
b) Explain cellular respiration and provide its chemical equation.
c) Difference between arteries and veins.
d) Adaptations of aquatic animals: give two examples.
e) Define food chain and give an example.

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2. Physics

a) A car travels 180 m in 15 s. Calculate its average speed.
b) Define potential energy and give one example.
c) Difference between radiation and convection.
d) A light ray strikes a mirror at 45°. Find the angle of incidence and reflection.
e) State and explain Newton's second law of motion.

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3. Chemistry

a) Identify as element, compound, or mixture: Carbon, H_2O, Soil.
b) Write a balanced chemical equation for the reaction of iron with oxygen.
c) Difference between polar and non-polar covalent bonds.
d) Define base and give an example.
e) Explain why mass is conserved in chemical reactions.

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4. Environmental Science

a) Name two greenhouse gases and their effects.
b) Suggest three ways to reduce energy consumption.
c) Explain acid rain and its impact on the environment.
d) How can reforestation help the ecosystem?

Section D: Social Studies

1. History

a) Explain the causes of the French Revolution.
b) Discuss the impact of the Industrial Revolution on society.
c) List major countries in the Axis powers during WWII.

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2. Geography

a) Define longitude and its importance.
b) Name the longest river in Asia.
c) Discuss two effects of urbanization on the environment.

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3. Civics / Political Science

a) Explain the difference between parliamentary and presidential systems.
b) State two rights of citizens in a democracy.

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4. Economics

a) Explain how price affects supply and demand.
b) Difference between revenue and profit.
c) Importance of saving and investment in personal finance.

Section E: ICT / Computer Studies

1. Computer Basics

a) Define LAN and WAN with examples.
b) Name one input and one storage device.

2. Applications

a) Which spreadsheet function finds the minimum value?
b) Two advantages of using word processing software in schools.

3. Programming Basics

a) Define conditional statements with one example in Python.
b) Write Python code to print even numbers from 2 to 20.

4. Cyber Awareness

a) Explain phishing and one way to prevent it.
b) List two tips to create strong passwords.

Section F: Islamiyat

1. Quran & Hadith

a) Name two values promoted by the Quran.
b) Narrate a Hadith emphasizing patience.

2. Aqidah / Belief

a) List the six pillars of Iman.
b) Explain the concept of Risalah (Prophethood).

3. Fiqh / Worship

a) Importance of Zakat in Islamic society.
b) Explain the etiquette of Hajj.

4. Islamic Ethics & Morality

a) How does Islam promote truthfulness in communication?
b) Suggest two ways Islam encourages care for animals.

Section G: Arts & Music

1. Visual Arts

a) Explain shading techniques in sketching.
b) Name three complementary color pairs.

2. Music

a) Define tempo and dynamics in music.
b) Give one example of a wind instrument and one percussion instrument.

Section H: Health & Physical Education

1. Physical Fitness

a) Name two exercises for cardiovascular health.
b) Distinguish between recreational and competitive sports.

2. Health & Hygiene

a) Why is personal hygiene important in school?
b) Two practices to prevent infectious diseases.

3. Safety & Environment

a) One safety rule for laboratory experiments.
b) How to prevent road accidents?